

M004 Week 7 — English Worksheet

Topic: Auto-Building Spaces — Loops + Functions + Chat Commands · **Course:** Functions & Games ·

Time: about 45 minutes

This week one chat command builds a **whole space by itself**. A loop repeats a room piece, a function builds each piece, and the **number argument** decides how big it grows.

1 · Predict

```
on chat command "dungeon" with number rooms:
  repeat rooms times:
    build_room()
    move forward 6
```

You type **dungeon 3** in chat. How many rooms get built? What does **move forward 6** do between rooms?


```
function build_room():
  build_wall(5)
  build_wall(5)
  build_wall(5)
  build_wall(5)

on chat command "dungeon" with number rooms:
  repeat rooms times:
    build_room()
    move forward 6
```

`build_room` reuses `build_wall` four times. You type `dungeon 2` — how many walls get built in total?


```
on chat command "towers" with number count:
  repeat count times:
    build_tower(random number from 3 to 8)
    move forward 4
```

You type `towers 5`. Are all five towers the same height? Why does the row of towers feel random?

2 · Spot the Bug 🐛

Each block below is broken. Rewrite it so it works, then explain why the original was wrong and why your fix works.

Bug A — `dungeon 4` should build four rooms in a row. Right now all four rooms get built **in the same spot**, so it looks like only one room.

```
on chat command "dungeon" with number rooms:
  repeat rooms times:
    build_room()
```

Hint: after each room, the builder needs to move before building the next one.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The number you type should decide how many rooms get built. But `dungeon 5` and `dungeon 1` both build **one** room.

```
on chat command "dungeon" with number rooms:
  repeat 1 times:
    build_room()
    move forward 6
```

Hint: the argument `rooms` is never used.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — Each room is **6 blocks long**. The rooms should sit side by side, but every new room breaks the back wall of the room before it.

```
on chat command "dungeon" with number rooms:
  repeat rooms times:
    build_room()
    move forward 5
```

Hint: compare the room length with the move distance.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Switch to your homework world. Make a chat command with a number argument that auto-builds a space — a loop should repeat a room piece.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.